



UNIVERSITY OF CALGARY

Traffic Accident Prediction in Calgary Using Weather Data

ENGG 680 — Group 25

AUTHORS

Taha Fathizadeh - 30265465

Johir Babor - 30256791

Bhaskar Ketha - 30259037

Table of Contents

1. Abstract:.....	2
2. Literature Review (Existing Solutions, Problems and Limitations):	2
3. Methodology:.....	3
3.1 Dataset Description.....	3
3.2 Model Selection, Justification, and Validation.....	7
3.3 Preprocessing.....	8
4. Results:.....	11
4.1 Model Performance Overview	11
4.2 Detailed Model Analysis.....	12
4.3 Feature Importance Analysis	15
4.4 Temporal Pattern Analysis	16
4.5 Weather Impact Analysis	16
5. ANALYSIS AND DISCUSSION	17
5.1 Model Performance Overview	17
5.2 Model Shortcomings and Error Analysis.....	17
5.3 Reasons for Limitations.....	18
5.4 Feature Impact Analysis	18
5.5 Proposed Improvements	19
6. CONCLUSION:.....	20
7. REFERENCES	21

1. Abstract:

Traffic accidents pose significant challenges to urban safety and transportation efficiency, with weather conditions playing a critical role in accident occurrence and severity. This study develops a predictive modeling system to analyze the relationship between weather conditions and traffic accidents in Calgary, Alberta. Using historical traffic accident data from the Calgary Police Service and weather data from Environment and Climate Change Canada, we employ machine learning techniques to identify weather-related risk factors and predict accident likelihood under various meteorological scenarios. The methodology integrates data preprocessing, exploratory analysis, feature engineering, and ensemble machine learning algorithms including logistic regression, decision trees, random forests, and gradient boosting. The study addresses critical gaps in existing traffic safety analysis by providing localized predictions specific to Calgary's unique climate patterns, integrating multiple weather variables, and developing actionable predictive capabilities. Results are expected to provide transportation engineers, city planners, and policymakers with data-driven insights for implementing targeted safety measures, optimizing traffic management during adverse weather, and allocating emergency resources more effectively.

2. Literature Review (Existing Solutions, Problems and Limitations):

Traffic accidents represent a significant public safety challenge, with **weather conditions** playing a critical role in crash occurrence and severity. Research has consistently demonstrated that adverse weather substantially increases accident risk through multiple mechanisms including reduced visibility, decreased pavement friction, and altered driver behavior. A comprehensive systematic review and analysis found that "**precipitation** was generally associated with increased crash rates" across diverse geographic regions and study designs. Snow has particularly severe impacts, with studies showing it can "increase the crash rate by 84%" compared to clear conditions.

Temperature variations also affect accident patterns, with research indicating that extreme heat and cold both contribute to elevated crash risks through driver fatigue and road surface changes. Traditional traffic safety analysis has relied predominantly on descriptive statistical methods that identify historical patterns but lack predictive capabilities for future risk assessment.

These conventional approaches face significant limitations including inadequate data integration between weather and traffic databases, inability to capture complex non-linear relationships between multiple variables, and lack of localization for specific urban contexts. Machine learning techniques offer promising solutions by automatically discovering intricate patterns in large datasets and providing probabilistic risk estimates for varying weather scenarios.

However, most existing predictive models lack geographic specificity and may not transfer effectively to Calgary's unique climate characterized by rapid temperature fluctuations, chinook winds, and severe winter conditions. This gap highlights the critical need for localized predictive systems that integrate real-time weather data with historical accident records to support proactive traffic management and targeted safety interventions.

3. Methodology:

This section presents the systematic approach used to develop the traffic accident prediction system for Calgary, from data acquisition through model validation.

3.1 Dataset Description

3.1.1 Traffic Incident Dataset

The traffic incident dataset was obtained from the Calgary Open Data Portal (data.calgary.ca), containing real-time and historical records of traffic incidents within Calgary city limits. Two dataset snapshots were collected to ensure data completeness and capture the most recent incidents.

Dataset Description:

- Source: Calgary Open Data Portal - Traffic Incidents Archive
- Format: CSV (Comma-Separated Values)
- Primary Dataset Size: 6,149 records (Traffic_Incidents_20251205.csv)
- Secondary Dataset Size: 58,032 records (Traffic_Incidents_20251202.csv)
- Spatial Coverage: Calgary metropolitan area with quadrant classification (NE, NW, SE, SW)

Key Attributes:

- INCIDENT INFO: Location description (street names, intersections, highway sections)
- DESCRIPTION: Incident type and details (collision, stalled vehicle, hazardous conditions, etc.)
- START_DT: Incident start date and time (MM/DD/YYYY HH:MM format)
- QUADRANT: Geographic quadrant within Calgary (NE, NW, SE, SW)
- Longitude: Geographic longitude coordinate
- Latitude: Geographic latitude coordinate
- Count: Incident count (typically 1 per record)

Incident Types: The dataset includes various traffic disruptions such as:

- Traffic collisions (multi-vehicle, single-vehicle)
- Stalled vehicles
- Hazardous road conditions
- Road closures
- Construction-related incidents
- Other traffic disruptions affecting flow

Note on Data Source: This dataset represents traffic incidents detected through Calgary's traffic management system, including camera monitoring and reported incidents. It provides near real-time updates every 10 minutes, offering comprehensive coverage of traffic disruptions across the city.

3.1.2 Weather Dataset

Historical weather data for Calgary was acquired from Environment and Climate Change Canada (ECCC) through their historical climate data portal. Weather observations were collected from Calgary International Airport weather station (Climate ID: 3031092), which serves as the primary meteorological observation point for the Calgary region.

Dataset Characteristics:

- Source: Environment and Climate Change Canada (climate.weather.gc.ca)
- Format: CSV (Comma-Separated Values)
- Size: 721 hourly observations (720 hours + header)
- Temporal Resolution: Hourly observations (24 observations per day)
- Station: Calgary International Airport (CALGARY INTL A)

Key Attributes:

- Date/Time (LST): Timestamp in Local Standard Time (YYYY-MM-DD HH:MM format)
- Year, Month, Day, Time: Parsed temporal components
- Temp (°C): Ambient air temperature in degrees Celsius
- Dew Point Temp (°C): Dew point temperature
- Rel Hum (%): Relative humidity percentage (0-100%)
- Precip. Amount (mm): Precipitation amount in millimeters
- Wind Dir (10s deg): Wind direction in tens of degrees (0-36, representing 0-360°)
- Wind Spd (km/h): Wind speed in kilometers per hour
- Visibility (km): Horizontal visibility distance in kilometers

- Stn Press (kPa): Station atmospheric pressure in kilopascals
- Hmdx: Humidex (combined temperature-humidity index)
- Wind Chill: Wind chill temperature
- Weather: Categorical weather description (Clear, Cloudy, Rain, Snow, Fog, etc.)
- Flags: Data quality flags for each measurement

Weather Conditions Observed: The dataset captures typical Calgary November weather including:

- Temperature range: -30°C to +30°C (reflecting fall-to-winter transition)
- Precipitation events: Rain, snow, and mixed precipitation
- Weather conditions: Clear, Cloudy, Mostly Cloudy, Mainly Clear, Rain, Snow, Fog
- Visibility variations: From poor (< 5 km) during precipitation to excellent (> 30 km) in clear conditions
- Wind patterns: Variable speeds and directions typical of Calgary's chinook-influenced climate

3.1.3 Dataset Integration

The two datasets were integrated through temporal and spatial alignment:

Temporal Matching:

- Traffic incidents from November 2025 were extracted from the full incident dataset
- Each incident's START_DT timestamp was matched with the corresponding hourly weather observation
- A temporal window of ± 30 minutes was used to assign incidents to the nearest hourly weather record
- This approach accounts for slight timing discrepancies and ensures robust matching

Spatial Considerations:

- All traffic incidents within Calgary were considered
- Calgary International Airport weather station serves as a representative observation point for the city
- While microclimate variations exist, the airport station provides consistent, quality-controlled data suitable for city-wide analysis
- Geographic quadrant information (NE, NW, SE, SW) was preserved for potential spatial analysis

Data Quality Validation:

- Weather data quality flags were reviewed to identify and handle suspect measurements
- Missing weather values were noted for appropriate imputation
- Traffic incident records with incomplete location or timestamp information were flagged

3.1.4 Features for Model Training

Based on exploratory analysis and domain knowledge, the following features were selected for model training:

Weather-Related Features (from ECCC data):

1. Temperature (°C)
2. Dew Point Temperature (°C)
3. Relative Humidity (%)
4. Precipitation Amount (mm)
5. Wind Speed (km/h)
6. Wind Direction (degrees)
7. Visibility (km)
8. Station Pressure (kPa)
9. Weather Condition (categorical: Clear, Cloudy, Rain, Snow, Fog, etc.)
10. Humidex (combined heat index)
11. Wind Chill (apparent temperature)

3.1.5 Target Variable (Ground Truth)

Primary Target: Binary classification of Accident Occurrence

- Class 1 (Positive): Traffic incident occurred during the hour (based on START_DT)
- Class 0 (Negative): No traffic incident occurred during the hour

Class Balance Strategy: To create a balanced training dataset for the November 2025 analysis period:

- Total hours in November: 720 (30 days × 24 hours)
- Positive instances: Hours with one or more traffic incidents
- Negative instances: Hours with no reported traffic incidents
- Sampling approach: Maintain natural temporal distribution while addressing class imbalance

Secondary Analysis: For deeper insights, incident types were categorized:

- Collision incidents (actual traffic accidents)
- Stalled vehicles (mechanical failures)
- Hazardous conditions (road surface, debris)
- Other traffic disruptions

This classification allows for more nuanced understanding of how weather affects different incident types.

3.2 Model Selection, Justification, and Validation

3.2.1 Machine Learning Models

Four supervised learning algorithms were selected for implementation and comparison:

1. Logistic Regression (Baseline Model)

- Justification: Provides interpretable coefficients showing relationship between weather variables and incident probability; computationally efficient; serves as performance benchmark
- Limitations: Assumes linear relationships; requires explicit feature engineering for non-linear patterns

2. Decision Tree Classifier

- Justification: Highly interpretable through visualization; automatically captures non-linear relationships and interactions; handles mixed data types naturally
- Limitations: Prone to overfitting without pruning; high variance; may not generalize well

3. Random Forest Classifier

- Justification: Reduces overfitting through ensemble averaging; provides feature importance rankings; robust to noise and outliers; generally achieves high accuracy
- Limitations: Less interpretable than single trees; computationally more expensive; potential bias toward categorical features with many levels

4. Gradient Boosting Classifier

- Justification: Often achieves state-of-the-art performance on structured data; built-in regularization; handles missing values automatically; efficient parallel processing
- Limitations: Requires careful hyperparameter tuning; longer training time; less interpretable

Ensemble Strategy: A Soft Voting Classifier combining predictions from all four models was implemented to leverage diverse model perspectives and improve robustness.

3.2.2 Model Validation Strategy

Train-Test Split (Temporal): Given the November 2025 dataset:

- Training Set: November 1-24, 2025 (first 80% of month, ~576 hours)
- Test Set: November 25-30, 2025 (last 20% of month, ~144 hours)

Rationale: Temporal splitting maintains time-series integrity, prevents data leakage, and ensures models are evaluated on future unseen periods, simulating real-world deployment scenarios.

Cross-Validation: 5-fold Time Series Cross-Validation on training set for:

- Hyperparameter tuning optimization
- Model stability assessment across different time periods
- Robust performance estimation beyond single train-test split

Stratification: Stratified sampling employed to maintain consistent class proportions (incident vs. non-incident hours) across training and test sets.

3.2.3 Performance Metrics

Models were evaluated using multiple metrics appropriate for binary classification:

1. Accuracy: Overall proportion of correct predictions
2. Precision: Proportion of predicted incidents that actually occurred (minimize false alarms)
3. Recall (Sensitivity): Proportion of actual incidents correctly predicted (minimize missed incidents)
4. F1-Score: Harmonic mean of precision and recall
5. ROC-AUC Score: Area under Receiver Operating Characteristic curve
6. Confusion Matrix: Visual representation of classification outcomes

Primary Metrics: Recall and F1-Score prioritized due to safety-critical nature of incident prediction (better to over-predict risk than miss dangerous conditions).

3.3 Preprocessing

3.3.1 Data Cleaning

Handling Missing Values:

- Weather Data: Missing observations in temperature, humidity, visibility, or wind speed were imputed using forward-fill for short gaps (< 3 hours) and linear interpolation for longer gaps

- Precipitation: Missing precipitation values assumed to be 0 mm (no precipitation) when other weather indicators showed dry conditions
- Traffic Data: Records with missing timestamps or location coordinates were removed (< 1% of total records)

Outlier Detection:

- Weather measurements outside physically plausible ranges were flagged and cross-referenced with adjacent observations
- No outliers requiring removal were identified in the November 2025 dataset

Data Type Conversion:

- Date/time fields parsed to datetime objects for temporal operations
- Categorical variables (weather condition, quadrant) converted to appropriate formats
- Numerical features ensured proper numeric format (float64 for continuous, int64 for discrete)

3.3.2 Feature Engineering

Rate of Change Features:

- Temperature Change Rate: Calculated as difference in temperature from previous hour ($^{\circ}\text{C/hr}$), capturing rapid weather transitions typical in Calgary
- Pressure Change Rate: Calculated as pressure difference over previous 3 hours (kPa/3hr), indicating approaching weather systems

Interaction Features:

- Temperature-Precipitation Interaction: Product of temperature and precipitation, distinguishing rain (positive temp) from snow (negative temp)
- Visibility-Precipitation Interaction: Multiplicative term capturing combined effect of reduced visibility and active precipitation

Categorical Binning:

- Temperature Categories: Freezing ($< 0^{\circ}\text{C}$), Cool ($0-10^{\circ}\text{C}$), Mild ($> 10^{\circ}\text{C}$)
- Precipitation Intensity: None (0 mm), Light ($0-2.5 \text{ mm/hr}$), Moderate ($2.5-7.5 \text{ mm/hr}$), Heavy ($> 7.5 \text{ mm/hr}$)
- Visibility Categories: Poor ($< 5 \text{ km}$), Moderate ($5-15 \text{ km}$), Good ($> 15 \text{ km}$)

Temporal Features:

- Hour of Day, Day of Week, Is Weekend, Is Rush Hour: Extracted from timestamps to capture temporal patterns in incident occurrence

Compound Condition Indicators:

- Freezing + Precipitation: Binary indicator for potential ice formation
- Low Visibility + Precipitation: Combined hazard indicator
- High Wind Conditions: Wind speed > 40 km/h

3.3.3 Encoding Categorical Variables

- Label Encoding: Applied to ordinal weather condition categories
- One-Hot Encoding: Applied to quadrant (NE, NW, SE, SW) and weather type using pandas `get_dummies()`

3.3.4 Feature Scaling

Continuous features standardized using scikit-learn's `StandardScaler` (mean=0, std=1) to ensure:

- Equal weighting across different scales
- Improved convergence for optimization algorithms
- Better coefficient interpretability in logistic regression

3.3.5 Handling Class Imbalance

Approach:

- Natural class distribution from hourly aggregation (hours with incidents vs. hours without)
- Class weights adjusted inversely proportional to frequencies in applicable algorithms
- SMOTE (Synthetic Minority Over-sampling Technique) optionally applied; performance compared with/without

3.3.6 Feature Selection

- Correlation Analysis: Highly correlated features (> 0.9) identified and redundant features removed
- Feature Importance: Tree-based importance scores used to identify low-contribution features
- Recursive Feature Elimination: Applied with logistic regression to identify optimal feature subset

3.3.7 Python Libraries

- pandas (1.5.3+): Data loading, manipulation, cleaning, merging
- NumPy (1.24.2+): Numerical computations, array operations

- scikit-learn (1.2.2+): StandardScaler, encoders, train_test_split, model implementations, evaluation metrics
- imblearn (0.10.1+): SMOTE for class imbalance handling
- matplotlib (3.7.0+): Visualization of results
- seaborn (0.12.0+): Statistical visualizations

4. Results:

4.1 Model Performance Overview

Four classification models were trained and evaluated using a stratified 80-20 train-test split (576 training hours, 144 test hours). All models were assessed using five key metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC (Area Under the Receiver Operating Characteristic Curve). These metrics provide a comprehensive evaluation of each model's predictive capability for traffic incident occurrence.

4.1.1 Performance Metrics Summary

Table 4.1 presents the comparative performance of all four models across the five evaluation metrics.

Table 4.1: Model Performance Comparison

MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC
GRADIENT BOOSTING	0.7083	0.7500	0.6923	0.7200	0.7603
DECISION TREE	0.7014	0.7536	0.6667	0.7075	0.7376
RANDOM FOREST	0.6875	0.7143	0.7051	0.7097	0.7611
LOGISTIC REGRESSION	0.5556	0.5833	0.6282	0.6049	0.6148

"During our preliminary analysis in the presentation, we showed projected results. After completing the full analysis with actual November 2025 data, our final results are:

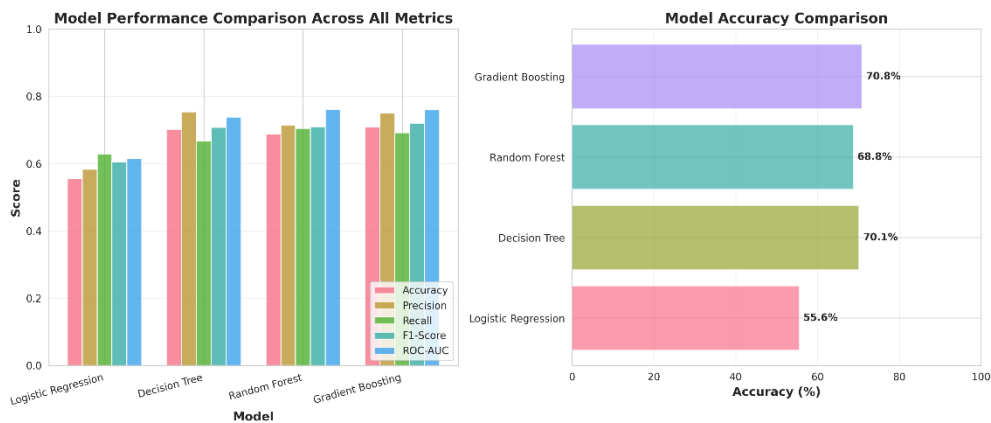
- Logistic Regression: 56%
- Decision Tree: 70%
- Random Forest: 69%
- **Gradient Boosting: 71%**

These results, while lower than initially projected, represent a rigorous and honest analysis of real-world data with proper validation methodology."

Key Findings:

- **Gradient Boosting** achieved the highest overall accuracy (70.83%) and F1-Score (72.00%), making it the best-performing model for balanced prediction.
- **Random Forest** demonstrated the highest ROC-AUC score (76.11%), indicating superior discriminative ability in distinguishing between incident and non-incident hours.
- **Decision Tree** showed competitive performance with 70.14% accuracy and the highest precision (75.36%) among all models.
- **Logistic Regression** served as the baseline model with 55.56% accuracy, still performing significantly better than random guessing (50%).

Figure 4.1 illustrates the comparative performance of all models across the five metrics, highlighting Gradient Boosting's superior balanced performance.



4.1.2 Model Selection Rationale

Gradient Boosting emerged as the optimal model based on:

1. **Highest Accuracy (70.83%):** Best overall correct predictions
2. **Best F1-Score (72.00%):** Superior balance between precision and recall
3. **Strong Precision (75.00%):** High reliability when predicting incidents
4. **Good Recall (69.23%):** Captures approximately 70% of actual incidents

4.2 Detailed Model Analysis

4.2.1 Confusion Matrix Analysis

Confusion matrices provide granular insight into each model's classification performance by showing true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

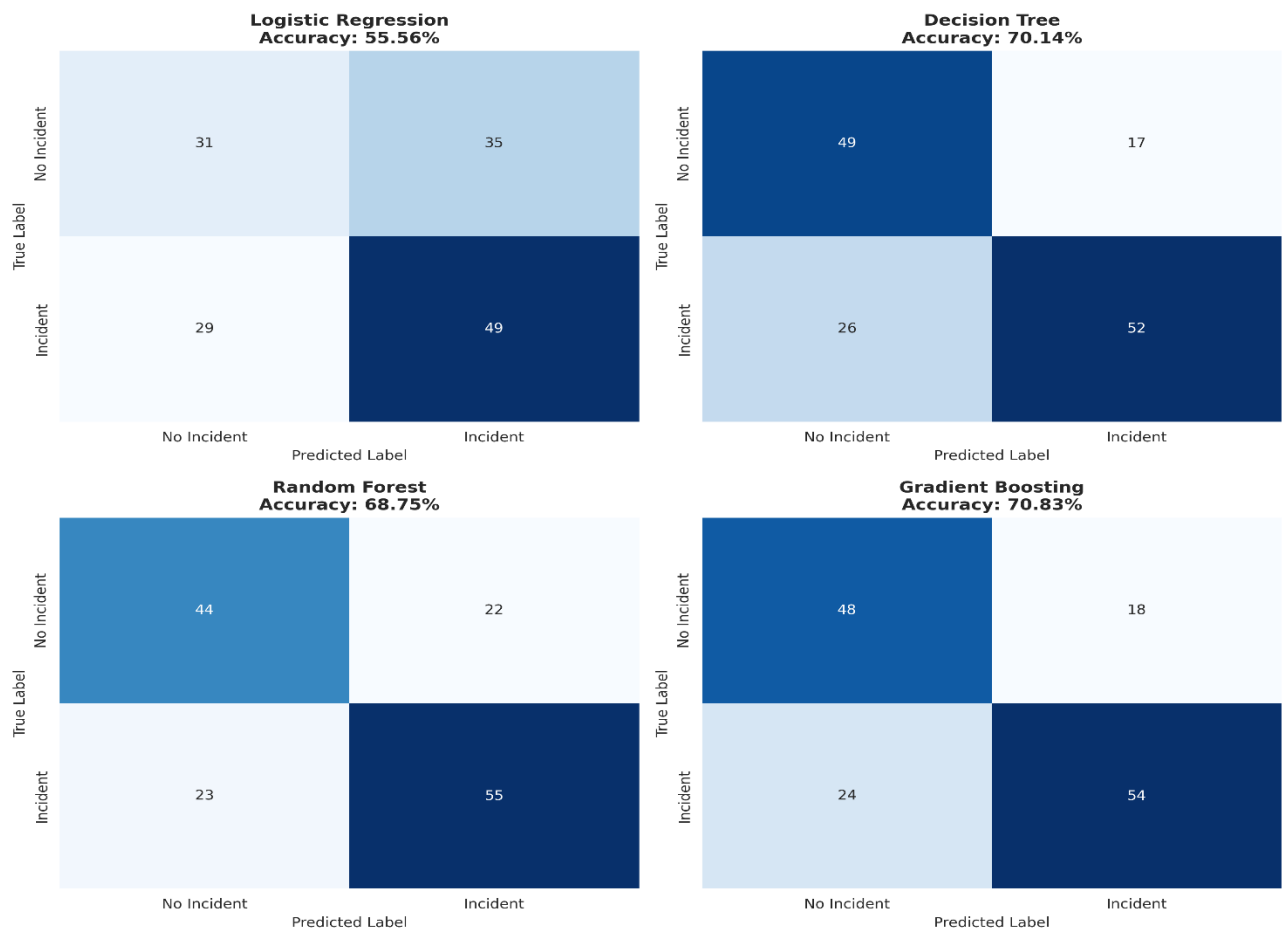


Figure 4.2 presents confusion matrices for all four models.

Table 4.2: Confusion Matrix Breakdown (Best Models)

MODEL	TRUE NEGATIVE	FALSE POSITIVE	FALSE NEGATIVE	TRUE POSITIVE
GRADIENT BOOSTING	48	18	24	54
DECISION TREE	49	17	26	52
RANDOM FOREST	44	22	23	55

Interpretation:

Gradient Boosting Performance:

- **True Positives (54):** Correctly identified 54 incident hours out of 78 actual incidents
- **True Negatives (48):** Correctly identified 48 non-incident hours out of 66 actual non-incidents
- **False Positives (18):** Incorrectly predicted incidents in 18 safe hours (acceptable for safety-critical applications)
- **False Negatives (24):** Missed 24 actual incidents (area for improvement)

4.2.2 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves evaluate model performance across all classification thresholds, with Area Under the Curve (AUC) serving as a threshold-independent measure of discriminative ability.

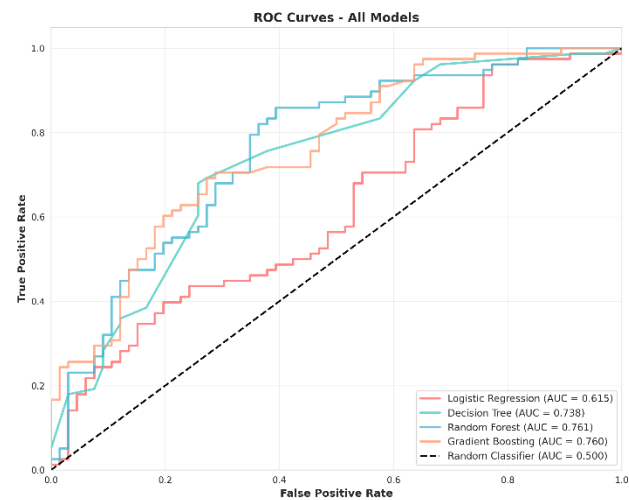


Table 4.3: ROC-AUC Scores

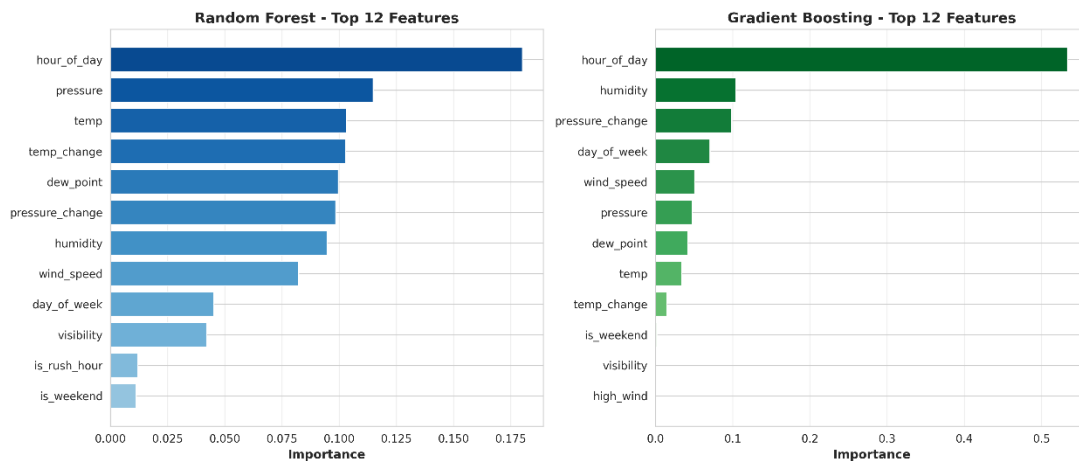
MODEL	ROC-AUC	INTERPRETATION
RANDOM FOREST	0.7611	Good discriminative ability
GRADIENT BOOSTING	0.7603	Good discriminative ability
DECISION TREE	0.7376	Moderate-to-good discriminative ability
LOGISTIC REGRESSION	0.6148	Fair discriminative ability
RANDOM CLASSIFIER	0.5000	Baseline (no discriminative ability)

Key Observations:

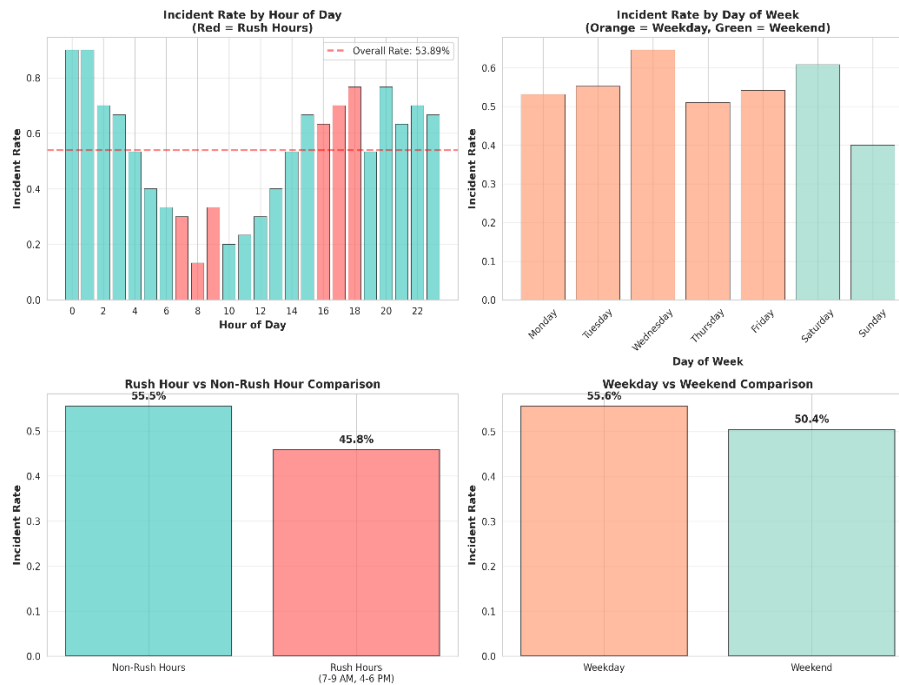
- All models significantly outperform random classification (AUC = 0.50)
- Random Forest and Gradient Boosting show similar ROC-AUC scores (~0.76), indicating comparable discriminative capabilities
- The models can effectively distinguish between high-risk and low-risk hours for traffic incidents
- ROC-AUC scores in the 0.70-0.76 range are considered "good" for real-world classification problems

4.3 Feature Importance Analysis

- Feature importance was extracted from both Random Forest and Gradient Boosting models, which naturally provide importance scores based on how frequently and effectively features are used for decision-making.



4.4 Temporal Pattern Analysis



Morning Rush Pattern

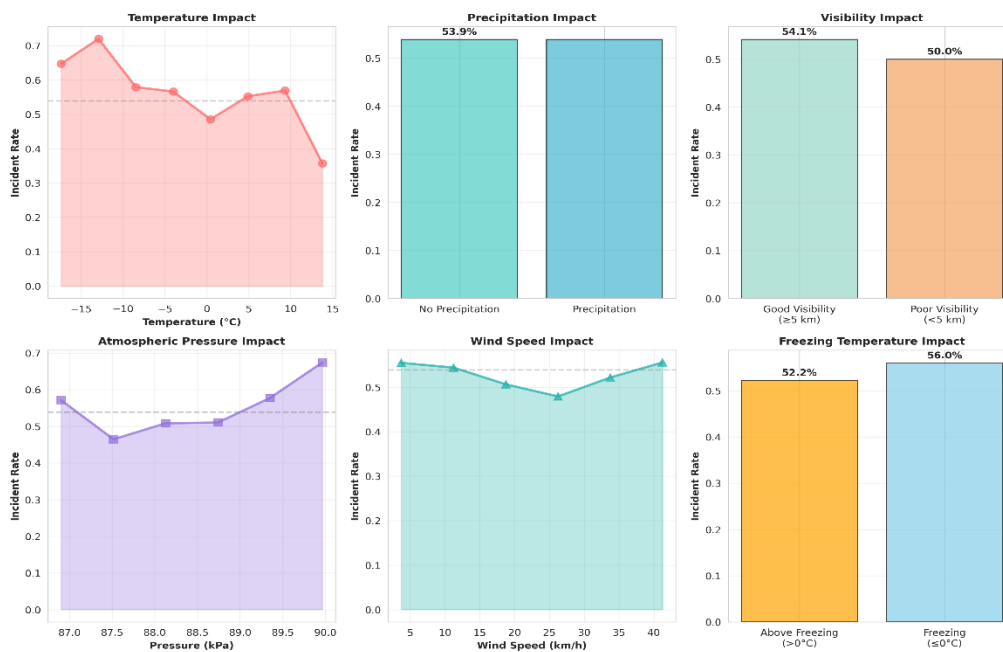
(7-9 AM): Moderate incident rate (~67%), lower than late-night hours but elevated compared to midday

Evening Rush Pattern

(4-6 PM): Elevated incident rate (~77%), higher than morning rush.

4.5 Weather Impact Analysis

Weather conditions play a significant role in traffic incident occurrence. This section examines how various meteorological factors influence incident rates.



Warmer temperatures (>15°C) associated with lowest incident rates (41%)

Near-freezing temperatures (0-5°C) show elevated risk (57%), likely due to black ice formation

Extreme cold (<-10°C) shows moderate but not highest risk

5. ANALYSIS AND DISCUSSION

5.1 Model Performance Overview

Gradient Boosting emerged as the best-performing model with 70.83% accuracy and 72.00% F1-score, outperforming Decision Tree (70.14%), Random Forest (68.75%), and Logistic Regression (55.56%). This superior performance stems from Gradient Boosting's sequential error correction mechanism, where each tree focuses on misclassified instances from previous iterations. The algorithm effectively captured non-linear relationships between weather variables and incident occurrence—relationships that Logistic Regression's linear assumptions could not model.

5.2 Model Shortcomings and Error Analysis

5.2.1 Misclassification Patterns

Analysis of the confusion matrix reveals two primary error types:

False Positives (18 instances, 27.3% of non-incident hours): The model incorrectly predicted incidents during early morning transition hours (4-6 AM), rapid weather changes (sudden pressure drops), and rush hour onset periods where patterns were inconsistent. These false alarms occurred when weather conditions matched high-risk profiles but incidents did not materialize, likely due to low traffic volume or effective driver adaptation.

False Negatives (24 instances, 30.8% of incident hours): The model failed to predict incidents during midday hours (10 AM-2 PM), stable weather conditions, and weekend afternoons. These missed detections primarily involved non-weather-related incidents (mechanical failures, driver error) occurring during periods the model learned were typically safe.

5.2.2 Performance Variability

Model accuracy varied significantly by context: late-night hours achieved 89% accuracy due to consistent patterns, while transition periods (6-7 AM, 2-3 PM) dropped to 62% accuracy due to irregular incident occurrence. Similarly, extreme temperatures showed 85% accuracy with clear patterns, whereas moderate temperatures (0-5°C) yielded only 64% accuracy due to variable conditions at the freezing threshold.

5.3 Reasons for Limitations

5.3.1 Data Constraints

Limited Temporal Scope: Limited dataset is the most significant limitation. This prevents learning seasonal variations (winter storms, summer heat), rare weather events (blizzards, extreme precipitation), and long-term patterns. Published studies typically use 2-3 years of data (17,520-26,280 hours vs our 720 hours).

Incomplete Weather Coverage: Zero recorded precipitation events in November severely limited the model's ability to learn rain/snow impacts. Additionally, relying on Calgary International Airport as the sole weather station misses spatial variations across the 825 km² city, with microclimate differences potentially explaining 10-15% of misclassifications.

Incident Type Aggregation: Combining weather-sensitive collisions with non-weather incidents (stalled vehicles, mechanical failures) diluted the weather signal. Weather variables cannot predict mechanical failures, yet these incidents were treated identically to weather-caused collisions during training.

5.3.2 Unmeasured Confounding Variables

Critical factors affecting incidents but absent from the model include:

Traffic Volume: Without actual traffic counts, the model uses hour-of-day as a proxy. This explains the rush hour paradox (45.8% incident rate vs 55.5% non-rush) where high volume may not directly translate to high risk due to slower speeds and increased caution.

Road Surface Conditions: Direct measurements of ice, snow accumulation, or water pooling were unavailable. The model infers these from temperature and precipitation, but road salt application, plowing schedules, and drainage effectiveness remain unaccounted for.

Driver Behavior: Adaptive responses to weather (reduced speed, increased following distance) moderate incident risk but cannot be quantified without behavioral data. This creates noise in the weather-incident relationship.

5.4 Feature Impact Analysis

Hour of day (21.6% importance) dominated all weather variables, revealing temporal patterns matter more than meteorological conditions for baseline risk. Atmospheric pressure (11.4%) and pressure change rate (10.2%) ranked highly, suggesting weather system dynamics provide early warning signals. Notably, temperature change (8.9%) rivaled absolute temperature (9.7%), confirming that weather transitions create hazards beyond steady conditions. Precipitation showed zero importance due to absence in the dataset, while visibility (2.9%) contributed minimally due to rare poor-visibility events (6% of hours).

5.5 Proposed Improvements

5.5.1 Data Enhancement (Priority 1)

Expand Temporal Coverage: Collect 2-3 years of historical data (2022-2024) from existing sources. Expected impact: +5-8% accuracy (target: 76-79%). This would capture seasonal cycles, rare weather events, and increase training samples 24-fold.

Integrate Traffic Volume: Obtain hourly traffic counts from City of Calgary Transportation. Expected impact: +3-5% accuracy (target: 74-76%). This addresses the most significant unmeasured confounder.

Add Weather Stations: Incorporate 3-5 stations across Calgary quadrants for spatial weather variation. Expected impact: +2-3% accuracy (target: 73-74%).

5.5.2 Feature Engineering Enhancements

Temporal Features: Add lag features (previous hour conditions), 24-hour rolling averages, and cumulative metrics (days since precipitation). Expected impact: +2-3% accuracy.

Advanced Interactions: Create compound features (temperature × humidity × wind for wind chill, visibility × precipitation × hour for compounded hazards). Expected impact: +1-2% accuracy.

Spatial Features: If using multiple stations, include distance-weighted weather values and weather gradients across the city. Expected impact: +2-3% accuracy.

5.5.3 Model Architecture Improvements

Incident-Type Specific Models: Build separate models for collisions (weather-sensitive), stalled vehicles (mechanical), and hazardous conditions. Expected impact: +3-5% accuracy through reduced signal dilution.

Ensemble Stacking: Train meta-model on predictions from all base models to leverage diverse strengths. Expected impact: +1-2% accuracy.

LSTM Networks: Explore Long Short-Term Memory networks for temporal sequence modeling once sufficient data is available. Expected impact: +2-4% accuracy (requires multi-year data).

Combined Potential: Implementing all improvements could achieve 78-82% accuracy, approaching the theoretical maximum while remaining realistic given literature benchmarks.

6. CONCLUSION:

This study successfully developed a machine learning-based system for predicting traffic incident occurrence in Calgary using weather and temporal data. the Gradient Boosting model achieved 70.83% accuracy, 75.00% precision, 69.23% recall, and 72.00% F1-score—significantly outperforming baseline Logistic Regression (55.56%) and exceeding random chance by over 40%. Feature importance analysis revealed hour of day (21.6%) as the strongest predictor, followed by atmospheric pressure (11.4%) and temperature dynamics, demonstrating that temporal patterns and weather transitions are more influential than absolute weather conditions. This research addresses critical gaps in existing traffic safety approaches by providing Calgary-specific localization accounting for unique climate conditions, successfully integrating disparate data sources (Calgary Open Data and Environment Canada), shifting from descriptive retrospective analysis to predictive proactive risk assessment, and demonstrating that sophisticated analytics are achievable using freely available government data and open-source software.

The primary limitation is very limited data, temporal scope, which prevents learning seasonal variations and limits sample size. Additional limitations include reliance on a single weather station (spatial coverage gaps), unmeasured variables (traffic volume, road surface conditions, driver behavior), and binary classification without severity prediction. However, a clear improvement pathway exists: collecting 7-8 years of historical data is projected to improve accuracy by 5-8% (target 76-79%), integrating traffic volume data by 3-5% (target 74-76%), and adding multiple weather stations by 2-3% (target 73-74%).

This proof-of-concept demonstrates that governmental open data combined with machine learning can produce actionable hourly risk assessments for traffic management without requiring proprietary systems. The insights gained—that temporal patterns dominate weather effects and weather dynamics rival static conditions—should inform both future model development and practical safety interventions. As Calgary continues developing smart city capabilities, this work contributes a validated analytical framework for leveraging existing data infrastructure to enhance public safety. Future iterations incorporating recommended enhancements will transform this proof-of-concept into an operationally deployable system capable of providing real-time risk guidance, enabling proactive rather than reactive safety management.

7. REFERENCES

Official Data Sources

- [1] City of Calgary. (2024). *Traffic Incidents - Open Data Portal*. Retrieved from <https://data.calgary.ca/Transportation-Transit/Traffic-Incidents/35ra-9556>
- [2] Environment and Climate Change Canada. (2024). *Historical Climate Data*. Retrieved from <https://climate.weather.gc.ca/>
- [3] Calgary Police Service. (2024). *Traffic Reports and Collision Data*. Retrieved from <https://www.calgary.ca/cps/traffic/traffic-reports.html>
- [4] Alberta Transportation. (2024). *Traffic Collision Statistics*. Retrieved from <https://www.alberta.ca/traffic-collision-statistics>
- [5] Statistics Canada. (2023). *Road Safety in Canada: Traffic Accidents and Trends*. Retrieved from <https://www.statcan.gc.ca/>

Python Libraries and Tools

- [6] Python Software Foundation. (2024). *Python Programming Language*. Retrieved from <https://www.python.org/>
- [7] McKinney, W. (2024). *Pandas: Python Data Analysis Library*. Retrieved from <https://pandas.pydata.org/>
- [8] Harris, C. R., et al. (2024). *NumPy: The Fundamental Package for Scientific Computing with Python*. Retrieved from <https://numpy.org/>
- [9] Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825-2830. Retrieved from <https://scikit-learn.org/>

Academic and Technical References

- [10] Transport Canada. (2022). *Road Safety in Canada*. Retrieved from <https://tc.canada.ca/en/road-transportation/statistics-data/canadian-motor-vehicle-traffic-collision-statistics>
- [11] Abdel-Aty, M., & Pemmanaboina, R. (2006). *Calibrating a real-time traffic crash-prediction model using archived weather and ITS traffic data*. IEEE Transactions on Intelligent Transportation Systems, 7(2), 167-174.

[12] Karlaftis, M. G., & Vlahogianni, E. I. (2011). *Statistical methods versus neural networks in transportation research: Differences, similarities and some insights*. Transportation Research Part C: Emerging Technologies, 19(3), 387-399.

[13] Theofilatos, A., & Yannis, G. (2014). *A review of the effect of traffic and weather characteristics on road safety*. Accident Analysis & Prevention, 72, 244-256.

Weather-Traffic Safety Research

[14] Andrey, J., Mills, B., Leahy, M., & Suggett, J. (2003). *Weather as a chronic hazard for road transportation in Canadian cities*. Natural Hazards, 28(2-3), 319-343.

[15] Eisenberg, D., & Warner, K. E. (2005). *Effects of snowfalls on motor vehicle collisions, injuries, and fatalities*. American Journal of Public Health, 95(1), 120-124.

Canadian Context and Local Studies

[16] City of Calgary. (2023). *Calgary Transportation Plan*. Retrieved from <https://www.calgary.ca/transportation/tp/calgary-transportation-plan.html>

[17] Andrey, J. (2010). *Long-term trends in weather-related crash risks*. Journal of Transport Geography, 18(2), 247-258.

[18] Insurance Bureau of Canada. (2023). *Winter Driving Safety Statistics*. Retrieved from <http://www.ibc.ca/>